

William Henry Melvin

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EDUCATION

James Madison University — Harrisonburg, VA

Bachelor of Science, Integrated Science & Technology

Expected Graduation: May 2028 | GPA: 3.1

Relevant Coursework: Thermodynamics, Applied Physics, Applied Statistics, Sustainable Manufacturing Systems, Programming, Biotechnology, Energy Issues, Calculus I & II

ENGINEERING EXPERIENCE

Engineering Intern — Ryzing Technology | Staunton, VA

May 2026 – August 2026

- Supported engineering projects for the U.S. Navy, U.S. Army, and NASA.
- Designed components in Onshape and created manufacturing-ready engineering drawings.
- Operated manual CNC equipment and other machining tools to manufacture prototype and production components.
- Performed ROI analysis for manufacturing equipment purchases.
- Participated in engineering design reviews and cross-functional project meetings.

Lifeguard

- Ensured patron safety through proactive monitoring and emergency response.
- Maintained CPR/AED certification and delivered excellent customer service.

ENGINEERING PROJECTS

Electronic Throttle Control System with Closed-Loop Speed Feedback

- Designed an Arduino-based electronic throttle control system.
- Integrated Hall effect sensors and tachometers to monitor motor speed and provide feedback.
- Redesigned the mechanism from a rigid coupler to a pulley-and-cable system for smoother throttle actuation.
- Designed custom CAD components and iterated prototypes.
- Troubleshooted electrical, mechanical, and software systems.
- Developing embedded motion controller with TFT interface and SD-card CSV motion profiles.

TECHNICAL SKILLS

CAD: SolidWorks, Onshape, Fusion 360

Manufacturing: Manual CNC, Manual Lathe, TIG Welding, 3D Printing, Engineering Drawings

Software: Microsoft Excel, Word, PowerPoint, Microsoft 365

CERTIFICATIONS

Google Project Management • Microsoft 365 Fundamentals • Microsoft Excel • Microsoft Word • Microsoft PowerPoint
• CPR/AED for Professional Rescuers and First Aid • Onshape Certifications